Shashwat Raj

(602)-471 6925 rajayshashwat@gmail.com raj-shashwat darthvader58 shashwatraj.com *Tempe, Arizona*

Education

Arizona State University **2023-2027**
B.S.E. in Computer Systems Engineering + B.S. in Mathematics (Dual Major), ***GPA : 3.8/4.0*** *Tempe, Arizona*

Activities & Leadership : GCSP Student Rep for ASU, former Technical Director at ACM@ASU, DevilQuant, Hacker Devils, Venture Devils, EPICS@ASU Project Leader, Engineering Futures Mentor, Section Leader for ASU 101 F24

Academic Awards & Scholarships : Dean's List F23-F25; NAmU Scholarship $13500/yr; SP25 Go-Global Scholarship $3000

Experience

Collective Design (CoDe) lab **May 2024 – Present**
Machine Learning Developer and Researcher *Tempe, Arizona*

- Developing **Reinforcement Learning** techniques to optimize Earth science missions to autonomously determine priority observations, under **Dr. Paul Grogan**. Co-**authoring** a review paper on intersection of **OSSEs** & **Mission Engineering**.
- Trained **DQN** & **QRDQN** models using **PyTorch**, GeoPandas, TAT-C on **NASA's G5NR dataset**, achieving **67.00% precision** & **87.00% recall**. Receiving total **$6200.00** through FURI and GCSP Research funding.

MentorU **July 2025 – August 2025**
Product Development Manager *Los Angeles, California*

- **Led** a **team** of 5-10 developers developing a **full-stack online platform** for college admission counseling startup, to automate features like scholarship finder and personal story-building. Increased UX Research success by **150%.**

WoofCare Solutions Pvt. Ltd. **May 2025 – Present**
Founder *New Delhi, India*

- Developing a mobile app using **Flutter** & **Firebase**, connecting dog lovers & dog care services to improve lives of stray dogs in India. Project funded by **EPICS@ASU** and partnered with over **60+ local and large NGOs** in Pune, India.

Projects

AI Privacy Firewall | *FastAPI, React, TypeScript, Gemma 4, Supabase, Transformers, D3.js, DistilBert* GitHub

- Asclepius is a clinical **AI privacy platform** that uses **neural-guided semantic proxy** to let pharma & clinical teams safely query latest remote **LLMs** like **Anthropic Claude Opus** and **OpenAI GPT5.5** without exposing **sensitive data**.
- It strips **18 HIPAA identifiers**, routes prompts through **3 privacy paths**, enforces a (ϵ=3.0, δ=1e-5) **DP budget**, and rehydrates answers locally so **zero raw PHI/MNPI** reaches the cloud; achieved **86% task utility**, tested **5 adversarial attack classes** on **50 synthetic clinical documents**, and won **HackPrinceton S26 Overall** with **$1150.00** in prizes.

Game Engine on FPGA | *SystemVerilog, Vivado, Xilinx, Synchronous Design, Hardware Synthesis, FSM* GitHub

- Built a **SystemVerilog FPGA** implementation of Dr. Robotnik's Mean Bean Machine on a **Nexys A7-100T**, driving **640**$\times$**480 VGA** at **60 Hz** with a **25 MHz pixel clock**, **4-bit RGB444** graphics, a **6**$\times$**12 real-time** game grid, **BFS**-based 4+ group clearing, gravity chains, **LFSR random generation**, and 6-digit seven-segment score display.

Kubernetes Cost Analyzer CLI | *kubectl, Go / GoLang, Bash, Cobra, Viper, clientcmd* GitHub

- KCAVO is a **kubectl plugin** that inspects live cluster state via **client-go**, projects monthly CPU, memory, GPU, and storage spend using a **730-hour billing model**, supports **JSON/YAML/table exports**, and flags waste patterns such as pods over **4 vCPU**, memory requests over **16GB**, GPUs above **70% of pod cost**, and **30-50% downscaling** savings opportunities.

F1 Algorithmic Strategy Game | *Next.js, Compiler Design, Pixi.js, PyDantic, Uvicorn, Docker, WebSockets* GitHub

- A multiplayer game where players write **Python pit-stop algorithms** competing in a **physics** based F1 **simulation**. Built a race engine, modeling tyre degradation cliffs, **Markov-chain weather**, safety cars, DRS, **sandboxed code execution**, **ELO matchmaking**, **6 adaptive AI bots**, and **real-time** race visualization using **8,500+ LOC** & **21 API** endpoints.

Other Projects and Publications listed on GitHub profile, LinkedIn and Portfolio Website

Technical Skills and Other Interests

Languages: Python, Java, C/C++, Typescript, Javascript, Rust, HTML/CSS, Dart, MySQL, LaTeX, React, Node.js, Ruby, CUDA, R, Matlab, Bash, Verilog, MIPS Assembly, Swift, C#, Visual Basic, Embedded C

Technologies/Frameworks: Django, Flask, Redis, OpenCV, Keras, Scikit Learn, Pandas, Git, GitHub, TensorFlow, Tailwind, Flutter, AWS, MongoDB, XGBoost, Hugging Face Transformers, NumPy, Docker, Matplotlib, Linux, KiCad, Arduino, ROS, Seaborn, Ansys, GraphQL, WebGL, OpenGL, LangChain, LangGraph, Agile Methods, Jira

Open-Source Contributions: Mocha JS, Electron JS, Pixi JS, Keras, Kubernetes, Supabase, OMI AI hardware

Side Interests: Competitive Programming, Hackathons, Algorithmic Trading, Datathons, Chess, Idea Pitching, Battlebots, Pololu Micromouse

Other Courses & Certifications listed on LinkedIn